Readable Is Not Used, Used Is Not Needed: An Inference-Only Audit of Continuous-Thought Reasoners

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Abstract

Some language models reason in a handful of hidden continuous vectors ("thoughts") instead of written steps. People want to monitor those thoughts for safety, but nobody has checked, on released checkpoints, which thoughts the model actually uses, which it cannot do without, and whether reading a thought tells you anything the model acted on. We built a one-command audit that answers these questions without any training, and ran it on five released checkpoints (CODI on GPT-2, CODI on LLaMA-3.2-1B, SIM-CoT on LLaMA-3.2-1B and 3B, and a Coconut GPT-2 retrain), four datasets, and 31 preregistered Kaggle runs. For the CODI family we find: (i) disturbing a thought and using a thought are different things, and below 3B a thought carries a value into the answer only on problems with three or more steps; (ii) exactly one or two thoughts do the carrying, they hand the value to the next position's memory, and below 3B the last thought never matters; (iii) the larger model uses its carrier thought but barely needs it, while the model's memory of the question is the single most necessary site; (iv) a probe on any thought, including the unused last one, reads the question well, so a monitor can look accurate while reading nothing the model computed; (v) the thoughts predict wrong answers no better than the model's own confidence; and (vi) a value-matched write into the carrier thought takes: transplanting a natural thought that holds the right intermediate value repairs 14–23% of failed multi-step problems against a 2–4% control, with 5–9% collateral damage at 1B and 3B. Scaling SIM-CoT to 3B removes the gate, puts the last thought to work and moves the carriers to other slots; it doubles the share of failures whose explicit chain is right (39%) but does not make them more writable, so thought correction stays at most a few points. The cheap gain is elsewhere: routing only the quarter of problems the latent pass is least confident about to the model's own explicit chain gains 3.7 points at 1B and 7.5 at 3B, and routing half beats running the chain everywhere. Coconut is the opposite case: its thoughts carry values on one-step problems, every pair is necessary (45–49 points), its last position is in use, and it cannot be written to. All results reproduce byte for byte across sessions. The audit costs about one GPU-hour per model and tells you, before you trust a monitor, which kind of model you have.

1 Introduction

A model that reasons in text can be watched: the steps are tokens, and a monitor can read them. Several recent methods instead train models to reason in a short run of continuous vectors that nobody can read directly, among them Coconut (Hao et al., 2024), CODI (Shen et al., 2025) and SIM-CoT (Wei et al., 2026). The usual response is to read the vectors with a probe (Ramjee et al.,

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2026). That response rests on three assumptions: that the probed vector holds something the model computed, that the model goes on to use it, and that the model would answer differently without it. None of these has been tested on the released checkpoints with a method that another lab could rerun.

This paper tests all three with interventions only. We never train or fine-tune a model. We take each checkpoint as released, swap, remove and read its thoughts, and measure what the answer does. Everything was preregistered in a running ledger before each run, scored against the prediction afterwards, and every number in this paper comes from a committed artifact.

Contributions.

1. **A method that separates three questions a probe conflates.** "Twin" problems that differ in one number tell us whether a thought *carries* a value (transport). Masking tells us whether the model *needs* it (necessity). Probes tell us whether it is *readable*, and the twins tell us whether what a probe reads is *confirmed by behaviour* (faithfulness). We also separate the two routes by which a thought can act: the state it feeds forward and the keys and values it leaves in memory.
2. **A standardised card and a comparison tool.** One command produces one card per (model, dataset, GPU session) with fixed thresholds, five checks, a regime and a monitor-placement verdict, plus fingerprints that let two sessions be compared bit for bit. We ran it on five models, four datasets and two Kaggle accounts.
3. **Findings about the CODI family** (Sections 5.1–5.3): a chain-length gate on transport and a trained carry-and-store ladder with an inert last thought, both of which the 3B model outgrows; sufficiency without necessity at 1B and 3B; the question register as the most necessary site; and readable-but-unfaithful thoughts under two probe families.
4. **Four uses of the map** (Sections 6.1–6.4): a prompt-leakage control that any latent-monitor evaluation should include; a negative result on error detection from thoughts; and a positive one, a value-matched latent repair by transplanting a natural thought that holds the right value, with controls that editing work so far has lacked, with a value from the answer key and, more weakly, from the model's own explicit chain plus a calculator; and a confidence-gated cascade that sends only the problems the latent pass is unsure about to that explicit chain, which beats every thought write at a fraction of the explicit cost.
5. **A contrast architecture** (Section 7): on Coconut the audit returns the opposite profile on almost every axis, which shows that the CODI results are properties of its training, not of continuous thoughts in general.

We also report two corrections to our own earlier readings and one to the literature: the released CODI-1B checkpoint scores 55.5% on GSM8K, which matches its paper's main figure of 55.6%; the 51.9% quoted by later papers is an ablation row.

2 Related work

Continuous-thought models. Coconut (Hao et al., 2024) feeds the last hidden state back as the next input embedding and replaces reasoning steps with such thoughts by a curriculum. CODI (Shen et al., 2025) distils an explicit-chain teacher into six thoughts through a projector, in one stage. SIM-CoT (Wei et al., 2026) adds a step-level auxiliary decoder during training. We study released CODI and SIM-CoT checkpoints and a third-party Coconut retrain, since no official Coconut GSM8K checkpoint exists.

Causal analyses of latent thoughts. Cywiński et al. (2025) patched single latent vectors of a self-trained CODI-1B with averages from prompts holding the same or different intermediate values, on twenty templates with three steps, and found that the third and fifth vectors store the two step values, that same-value patches leave accuracy unchanged, and that the last two vectors can be replaced by random ones at no cost; in our indexing those are slots 1 and 3 and the terminal slots, so our carriers and inert last thought confirm their placement on natural data. Dilgren and Wiegreffe (2026) decode intermediate results from the top-1 integer token of CODI and Coconut thoughts and show that mean-ablating all latents costs CODI-1B 25 points on GSM8k-Aug and nothing on logical datasets.

Liang and Pan (2026) show on sequential synthetic tasks that CODI forms faithful intermediate states only when the task is shallow and otherwise routes the input to the readout by a near-direct path. SCIT (Ding et al., 2026) swapped value-cache layers on same-template pairs. Li et al. (2026) overwrite single steps with zeros and report flip rates on the official checkpoints. Aswal et al. (2026) show that decodable patterns appear in controls without recurrence and call for causal tests. Jin et al. (2026) find accuracy retained when latent computation is removed. Our twins extend Cywiński et al. (2025) to natural data, five checkpoints and chains of three or more steps, add the two-route decomposition, per-slot necessity against per-slot transport, the chain-length gate and the monitor layer, and run everything from released weights.

Monitoring and probes. Ramjee et al. (2026) detect misaligned reasoning in a backdoored Coconut-style model by probing early states, with the trigger in the prompt. Ferreira et al. (2026) argue monitorability depends on access and task structure. That decodability is not causality is established elsewhere (Open Problems, 2025; Decodability, 2026; Single-position, 2026); we measure it per slot on latent reasoners.

Interventions and repair. Chang et al. (2026) improve accuracy by 0.4–1.8 points with training-free geometric edits of an early cached thought on CODI and Coconut at 3B and 8B, and transplant whole latent sequences between instances, with no value-matched donor, no different-value or random-donor control and no harm measurement; editing a thought is therefore published, and what we add is the value-matched write with its controls, its harm rate and its ceiling. Huang and Qi (2026) find synthetic steering of continuous thoughts does not reach the output. Hidden Error Awareness (2026) find that on explicit-CoT models probes read errors well but patching does not repair. LTO (2026) train a correctness classifier on latent states as a reward. We report a repair that aims a specific value through a specific slot using natural donor states, and the honest comparison against reading the explicit chain.

3 Setup

Models. Table 1 lists the checkpoints. All CODI-family models produce six thoughts after a begin-of-thought token; each thought's output state goes through a projector into the next input, and each thought also writes keys and values (K/V) into the cache that every later position attends to. Coconut has no projector: the thought *is* the last hidden state, fed back as the next input embedding.

Table 1: Checkpoints. Accuracies are our greedy GSM8K-test reproductions (1,319 problems).

Name	Base	Source	Paper	Ours
CODI GPT-2	GPT-2 (124M)	`zen-E/CODI-gpt2`	43.7	43.6
CODI-1B	LLaMA-3.2-1B-Instruct	`zen-E/CODI-llama3.2-1b-Instruct`	55.6	55.5
SIM-CoT	LLaMA-3.2-1B-Instruct	`internlm/SIM_COT-LLaMA3-CODI-1B`	56.1	55.7
SIM-CoT-3B	LLaMA-3.2-3B-Instruct	`internlm/SIM_COT-LLaMA3-CODI-3B`	62.3	61.9
Coconut	GPT-2 (124M)	`UdeSA/coconut-0.2B-gsm8k-aug` (retrain)	34.6	34.6

Datasets ("pools"). Every experiment needs problems whose solution is an equation chain, written as `«16-3-4=9» «9*2=18»`, because the twins recompute the chain. We use four pools: the GSM8k-Aug training split used by CODI itself (2,048 rows, plus 14,336 extra rows for long chains), the original GSM8K test set with its calculator annotations as the chain (1,206 usable rows, 854 with three or more steps), the original GSM8K train set (6,998 rows), and ASDiv (Miao et al., 2020) with its formula split into steps (1,490 rows, 86 with three or more steps). SVAMP (Patel et al., 2021) serves as a one-step control (300 rows). Problems with at most two steps are "short"; three or more are "long".

Outcome. For the CODI family we read the model's first answer token under the forced cue "The answer is:", which the released decoder uses, and also run native greedy decoding for headline conditions. Every rate comes with a 95% paired bootstrap interval over problem pairs (2,000 resamples). For Coconut, which has no forced cue, every outcome is native decoding.

4 Method

We describe the instruments in the order the audit runs them. The language is plain on purpose; the code is public and every step has a unit test.

4.1 Two ways a thought can act

A thought at position s affects the answer by two routes. Its *state* is the vector it hands to position $s+1$ through the projector. Its *K/V* are the keys and values it writes at its own cache position, which every later position reads through attention. We can edit either route alone: replace the state before it is projected (the "state" route), or overwrite the keys and values at that position after they are written (the "K/V" route), on any subset of layers. Replacing all six states *and* all six positions' K/V is identical to replacing the K/V alone, because the last state is discarded and everything a state does, it does through the K/V its successors write; we verified this identity exactly. Feeding a thought its own recorded state, or its own K/V, reproduces the original logits to the last bit, which is the test that the plumbing is right.

4.2 Twins: does a thought carry a value, or only disturb?

Swapping a thought from an unrelated problem breaks answers, but breaking is not carrying. To test carrying we build *twins*: for a problem, change one integer in the question and recompute the chain, so the twin's answer is the correct answer to the changed question. For example, "12 cucumbers" becomes "13 cucumbers" and the chain `«12*5=60» «54+60=114»` becomes `«13*5=65» «54+65=119»`. A number is eligible only if it appears exactly once in the question and once in the chain, is not itself a result, and the recomputed results stay non-negative integers. About half of GSM8k-Aug rows yield a twin.

Competence gate. We keep a pair only if the model solves both twins natively and at the first token, and their gold first tokens differ. Otherwise "the answer followed the swapped value" has no meaning.

Intervention and outcome. The two twins sit next to each other in the same batch, so their position ids match, and each is the other's donor. We replace one site in the recipient (one thought's state, or one position's K/V on a layer group, or whole tails) with the twin's and read the first answer token. The outcome is *target* if it becomes the twin's answer, *retain* if it stays the recipient's, *other* otherwise. The target rate is *transport*. Both directions are pooled.

Profiles. The *standard* profile draws 1,024 candidate twins from the pool in a fixed shuffled order and perturbs the first eligible number; most pairs are short chains. The *long* profile keeps only chains of three or more steps, perturbs the number that enters the chain latest, and draws from the extended pool. Both cap at 512 pairs. On the GSM8K test and train pools the caps are reached or the number of pairs is reported.

Locating the value. The model's own output head, applied to a thought's state, decodes a few numeric tokens. If the recipient's state at some thought decodes its own changed value and the twin's decodes the twin's, we say the value is *located* there. "Uniquely located" means at one thought only.

4.3 Masking: does the model need the thought?

Sufficiency is not necessity. For each thought we run two removals. *Mean*: replace its state by the average state over the pool, so nothing problem-specific reaches the next thought. *Mask*: hide its K/V from every later position through the attention mask, which needs a decoding path that carries the attention mask and explicit position ids through every step (the released LLaMA path drops the mask and we fixed that). We measure the accuracy drop in points, split by chain length, with intervals from a bootstrap over rows. Three preregistered checks: short chains survive any single removal within 5 points (K1), the carriers cost at least 10 points on long chains (K2), the last thought costs at most 2 points (K3).

4.4 What a probe is, and how it reads a thought

A thought is a vector of 768 numbers (GPT-2) or 2,048 numbers (the 1B models). A *probe* is a small model trained, after the fact, to predict a label from such a vector. It never touches the reasoning model; it only reads vectors we recorded. We use three readers.

1. **The model's own output head (no training).** The reasoning model ends with a matrix that turns a vector into scores over its vocabulary. Applying it to a thought's state asks "if this vector were about to be spoken, which token would it say?" We keep the numeric tokens and read the top few. This reader costs nothing and uses the model's own notion of a number; it is what the released decoder would do if it decoded the thought.
2. **A ridge probe (linear).** We fit a linear map W so that xW is close to the embedding of the target number's token, with a small penalty on W to avoid overfitting (ridge regression, closed form). To read, we map the state and pick the numeric token whose embedding is nearest. The target during training is the chain's penultimate intermediate result, the value the carriers hold.
3. **An MLP probe (non-linear).** A two-layer network (hidden width 256) trained to classify the state into one of about 900 numeric tokens, with weight decay, 300 full-batch steps and a fixed seed. It can read structure a linear map cannot.

Held-out reading. Probes are trained on four fifths of the recorded pairs and read the remaining fifth, five times over, with folds assigned by twin pair so a probe is never read on a problem whose twin it trained on. *Readability* is the held-out accuracy. A readability of 0.79 at thought 3 of CODI-1B means that, on four problems in five it had not seen, the probe named the right intermediate value from that one vector.

Why readability is not enough. A probe can read a value from a vector the model never uses, the way a bystander can read a note the recipient threw away. So we ask a second question with the twins. After a swap at thought s, the recipient holds the twin's state there. A monitor placed at s would read the twin's value and raise an alarm, "the model is working with value v' now". *Faithfulness* is the share of those alarms the behaviour confirms: how often the answer actually became the twin's. At CODI-1B thought 3 the probe reads at 0.79 and the alarm is confirmed 0.36 of the time; at thought 4 it reads at 0.84 and the alarm is confirmed 0.05 of the time. Both are readable; only one is a monitor.

Prompt-leakage probes. The same machinery, with a different label, tests what else a probe can read. We train probes to predict two things the question already fixes before any reasoning: the first operator in the chain and whether the problem has three or more steps. If these are readable from every thought, a probe's accuracy on a prompt-fixed property says nothing about the plan.

4.5 Monitors: readable, and faithful?

A monitor is a probe trained to read a value from a thought's state (Section 4.4). We train two families on the recorded twin states: a ridge map from the state to the embedding of the target number token, decoded by the nearest numeric token; and a two-layer MLP classifier over the numeric vocabulary. The target is the chain's penultimate intermediate result. Folds are assigned by pair so a twin never trains the probe that reads it. *Readability* is held-out accuracy. *Faithfulness* asks: when a monitor placed at this thought reads the twin's value from the twin's state after a swap (that is, raises an alarm), how often did the answer actually follow? A thought can be readable and never faithful.

4.6 The card

The driver runs the twins (both profiles), the masking audit, the two probe families and the bootstraps, and writes one card. Fixed thresholds: a *carrier* is a thought whose long-chain state transport is at least 0.15; a *store* is a position whose K/V transport is at least 0.15 while its state does not carry; the short-chain ceiling is 0.10; *necessary* means at least 10 points on long chains; a thought is *readable* at probe accuracy 0.30 and *readable-but-unfaithful* if its alarm is confirmed at most 5% of the time. Five checks: C1, no single site transports on short chains (position 0's K/V excepted, since it holds the question); C2, a carrier exists on long chains; C3, the last thought is inert, unfaithful and

unnecessary; C4, readability is dissociated from faithfulness; C5, every carrier is necessary. The *regime* is *critical path* if C5 holds, *backup route* if carriers exist but C5 fails, *no transport* if there is no carrier. A long profile with fewer than 50 gated pairs does not decide carriers. The card records the GPU, library versions, dataset hash, and a SHA-256 fingerprint of every twin run's predictions, and a comparison tool flags disagreements across pools and sessions.

4.7 Prompt-leakage probes

To test whether a probe reads the plan or the prompt, we train probes on every thought to predict two properties the question already fixes: the first operator of the chain (four classes) and whether the chain has three or more steps. We report balanced accuracy against chance (0.25 and 0.50) and compare the best slot with the carriers.

4.8 Error detection from thoughts

For every GSM8K test problem, right or wrong, we record the thoughts and the native answer, apply the model's own output head to each thought (top-1 numeric value, its probability, the entropy, and agreement with the answer and the other thoughts), and train logistic classifiers (five folds) to predict "the answer is wrong": carrier slots only, inert slots only, all slots, the answer's own first-token confidence, and a classifier on all raw state features. We report AUROC with bootstrap intervals.

4.9 Repair by transplanting a natural thought

The audit says the carrier thought holds the intermediate value and passes it on. Repair asks whether that channel can be used in reverse: if the model got a problem wrong, can we put the right intermediate value into the carrier and get the right answer out? We cannot write a number into a vector directly, since we do not know the code. But we can find a problem the model solved where the carrier already holds that number, and borrow its vector.

Worked example (a real case from the CODI-1B run). "The elevator in Jack's building supports a maximum load of 700 kg. An adult weighs an average of 80 kg. If Jack rides the elevator with 8 other adults, by how much will they have exceeded the maximum load?" The gold chain is `«8+1=9»` `«9*80=720»` `«720-700=20»`, so the penultimate intermediate is $v^* = 720$. CODI-1B answers 40, and its output head reads 640 at the carrier: it multiplied the 8 other adults by 80 and forgot Jack. In the bank of 2,048 training problems the output head reads 720 at the carrier of five problems. We take the one whose carrier vector is most similar to this problem's (cosine), copy that vector in at thought 3 while decoding, and read the new answer: 20, the gold answer, so the problem is *repaired*. With a random different-value donor the answer stays 40 (*unchanged*); an answer that is neither would be *changed but wrong*.

Steps. (1) Decode the 2,048 bank problems once, recording all six thoughts and their K/V; read each carrier with the output head to index the bank by value. (2) Decode every test problem with three or more steps; keep the wrongly answered ones (up to 512) and 128 correctly answered ones as controls. (3) For each target, find bank entries whose carrier reads v^*; pick the nearest. (4) Decode the target again under each condition and classify the answer. (5) Report rates with bootstrap intervals, on the *primary stratum* (wrong targets whose own carrier reads something other than v^* and that have a donor) and on all wrong targets with a donor.

Controls, and what each rules out. A *random same-value* donor rules out the nearest-neighbour choice mattering. The *centroid* of all same-value bank vectors is a synthetic vector in the style of Cywiński et al. (2025); it tests whether a natural single vector is needed. A *random different-value* donor is the key control: if it repaired as often, the repair would be a disturbance that happens to shake a wrong answer loose, not the value. The *store K/V* and *state plus K/V* conditions test the second route. The donor's *whole K/V tail* is a ceiling on transport, not a repair. The 128 correct controls measure *harm*: how often the same transplant, with v^* set to the control's own penultimate value, breaks a right answer.

Tool-supplied value. Every CODI checkpoint was also trained on explicit chains, so decoding the bare question makes it write its own equations. A calculator re-evaluates each one, propagating

corrected results into later equations that reference them; the corrected last result is the tool's v^*. We report, as the honest ceiling, how often the explicit chain's own answer is right on the same failures.

4.10 Coconut adapter

We reimplemented Coconut's forward pass with the same hooks: six latent passes where each input embedding is the previous last hidden state, a state hook that may replace a thought before it is fed on, a K/V hook at each position, K/V masks, and batched cached decoding with left padding. Feeding a thought its own state or K/V reproduces every output exactly, and batch size does not change outputs. All Coconut outcomes are native answers after the ### marker.

4.11 Preregistration and reproducibility

Every experiment was written into a ledger with its predictions before the run, and scored afterwards; eleven predictions were partly or wholly wrong and are reported as such. Runs executed on Kaggle T4 GPUs through the API from two accounts. Five reruns reproduced their twin predictions byte for byte. Code, cards, per-target outcomes and the ledger are in the public repository.

5 Results: the CODI family

5.1 Influence is not transport, and transport has a chain-length gate

Table 2 gives the central numbers. Swapping the whole K/V tail changes 60–75% of answers on all chains, yet on short chains no single thought moves more than 10% of answers to the twin's value. On long chains, one thought (GPT-2 and SIM-CoT: thoughts 1 and 3; CODI-1B: thought 3 alone) transports the value 0.20–0.36 of the time, and the K/V of the following position transport it at the same rate. This holds on the augmented training rows, the original test set and the original train set, with overlapping intervals.

Table 2: Transport (target rate) under a single-site swap, long chains. Columns are thought states $s_0..s_5$ and K/V positions $k_0..k_5$; the carrier/store pairs are bold. Short-chain transport never exceeds 0.10 at any single site in the 1B-and-smaller models; SIM-CoT-3B's thought 4 transports 0.23 on short chains. Pairs: 512 unless stated.

Model	Pool	s_0	s_1	s_2	s_3	s_4	s_5	k_0	k_1	k_2	k_3	k_4	k_5	kv_all
GPT-2	aug-train	.00	**.15**	.00	**.33**	.00	.00	.12	.00	**.16**	.00	**.30**	.00	.69
GPT-2	test (103 pairs)	.00	.13	.00	**.36**	.00	.00	.21	.00	.13	.00	**.33**	.00	.68
GPT-2	train-orig	.00	.14	.00	**.30**	.00	.00	.15	.00	.13	.00	**.29**	.00	.61
CODI-1B	aug-train	.04	.01	.01	**.34**	.04	.00	.04	.04	.01	.00	**.26**	.05	.67
CODI-1B	test (152 pairs)	.06	.01	.00	**.30**	.03	.00	.07	.06	.01	.00	**.24**	.03	.67
CODI-1B	train-orig	.06	.02	.00	**.35**	.03	.00	.07	.06	.02	.00	**.29**	.04	.65
SIM-CoT	aug-train	.04	**.19**	.04	**.25**	.04	.00	.13	.04	**.18**	.04	**.23**	.04	.73
SIM-CoT	test (168 pairs)	.03	**.22**	.05	**.20**	.04	.00	.22	.03	**.21**	.05	**.20**	.04	.72
SIM-CoT	train-orig	.03	**.21**	.05	**.23**	.05	.00	.17	.03	**.22**	.05	**.22**	.05	.66
SIM-CoT-3B	test (207 pairs)	.06	.04	**.27**	.02	**.16**	.00	.09	.06	.04	**.28**	.02	.15	.68

The pattern is a ladder: an odd thought computes and projects into the next even position, which stores the result in its K/V, where the answer decode reads it. The pairing is trained, not fixed: (1→2, 3→4) in GPT-2 and SIM-CoT, (3→4) alone in CODI-1B. The sixth thought's state is discarded by the architecture and its K/V transport is at most 0.05 everywhere. On the one-step control SVAMP, no site transports more than 0.045 and the card reports that nothing is monitorable. On ASDiv the short-chain ceiling holds on all three models; its multi-step subset gave 17–30 pairs, too few to decide carriers, though the point estimates (0.12–0.15 at thought 3) point the same way. At 3B the ladder keeps its shape and flips its parity: SIM-CoT-3B carries at thoughts 2 (0.27 [0.21, 0.32]) and 4 (0.16) and stores at thought 3's K/V (0.28), and thought 4 also transports 0.23 on short chains, so the chain-length gate is a property of the 1B-and-smaller checkpoints, not of the family. Its sixth thought's K/V transport 0.15 on both profiles: at 3B the terminal position is in use.

5.2 Used is not needed

Table 3 gives the masking costs. No single thought costs more than 1.3 points on short chains in any 1B-or-smaller model (K1 holds there); SIM-CoT-3B's thought 2 costs 2.8. On long chains the carriers cost GPT-2 6–20 points and SIM-CoT 4–12, but CODI-1B only 2–4 points, with upper bounds below 10 on every pool. CODI-1B uses thought 3 (transport 0.30–0.35) and does not need it: the *backup route* regime. The single most necessary site in GPT-2 and SIM-CoT is not a thought but position 0's K/V, which holds the question: hiding it costs 19–38 points (GPT-2) and 33–59 points (SIM-CoT) on long chains, and for SIM-CoT it alone transports 0.44–0.66 of short-chain twins. The last thought costs at most 1.9 points in the 1B-and-smaller models (K3 holds there); at 3B it costs 8.4 [5.1, 11.9], as much as a carrier. SIM-CoT-3B is also backup route: its carriers cost 6.5 and 8.9 points, the whole tail 44, and its question register 10.8, a third of the 1B figure. GPT-2's carrier cost straddles the 10-point line across pools (6.5 [2.4, 10.6] on the test set, 11.7 [7.7, 16.0] and 12.5 [7.2, 17.9] on the others); we do not read a regime change into overlapping intervals.

Table 3: Accuracy drop in points when one thought is removed (larger of mean-ablation and K/V mask), long chains; 95% bootstrap intervals over rows where available. "k_0" is the question register.

Model	Pool	k_0	1	2	3	4	5	all	Regime
GPT-2	aug-train	37.6	20.2 [14.4, 26.2]	12.2	12.5 [7.2, 17.9]	13.7	1.9	52.9	critical path
GPT-2	test	18.7	12.2 [7.9, 16.8]	9.2	6.5 [2.4, 10.6]	9.8	0.8	27.1	backup route
GPT-2	train-orig	29.5	18.3 [13.8, 23.2]	9.5	11.7 [7.7, 16.0]	12.9	1.1	37.0	critical path
CODI-1B	aug-train	3.4	2.7	1.5	2.7	4.6	0.8	38.8	backup route
CODI-1B	test	4.3	4.1	1.9	3.8 [1.3, 6.5]	4.3	1.1	28.5	backup route
CODI-1B	train-orig	3.2	7.2	3.2	2.3 [0.3, 4.3]	4.6	0.9	52.4	backup route
SIM-CoT	aug-train	58.6	10.6	11.4	9.1	8.4	3.0	76.0	backup route
SIM-CoT	test	32.5	9.8 [6.5, 13.3]	7.0	4.3 [2.2, 6.8]	5.7	1.9	44.2	backup route
SIM-CoT	train-orig	49.0	12.3 [8.3, 16.3]	11.7	9.2 [5.7, 13.2]	11.7	1.4	68.2	backup route
SIM-CoT-3B	test	10.8	5.7 [2.7, 8.9]	6.5 [3.3, 9.8]	4.1	8.9 [5.7, 12.5]	8.4 [5.1, 11.9]	44.2	backup route

5.3 Readable is not faithful

Table 4 compares what a probe reads with what the behaviour confirms. In every model the carriers are the only thoughts whose alarms are confirmed (0.17–0.40). In the 1B-and-smaller models thoughts 4 and 5 are read at 0.4–0.8 by the ridge probe and confirmed at 0.00–0.05; at 3B the readable-but-unconfirmed thought is the store, thought 3 (read 0.55 / 0.31, confirmed 0.02 / 0.03). The original check C4 ("the most readable slot is inert and readability anti-correlates with faithfulness") held for GPT-2 under the ridge probe but not under the MLP, and not for the 1B models: it is a probe-dependent statement and we withdraw it. What holds on every card under both families is C4b: a thought exists that a probe reads at ≥ 0.30 whose reading the behaviour confirms $\leq 5\%$ of the time, and the last thought is such a slot in every 1B-and-smaller model, the store in the 3B model. Carrier faithfulness moves by at most 0.04 between probe families: faithfulness belongs to the slot, not to the probe.

Table 4: Monitor readability (ridge / MLP) and faithfulness (ridge / MLP) per thought, long chains, penultimate intermediate as the target. Carriers in bold.

Model	Pool	0	1	2	3	4	5
Readability ridge / MLP							
GPT-2	aug-train	.09/.10	**.33/.49**	.46/.48	**.40/.58**	.54/.56	.42/.57
GPT-2	test	.05/.03	.32/.36	.41/.23	**.40/.42**	.51/.27	.36/.40
CODI-1B	aug-train	.27/.17	.15/.06	.36/.24	**.79/.75**	.84/.79	.81/.73
CODI-1B	test	.19/.15	.06/.06	.16/.08	**.59/.52**	.69/.62	.68/.48
SIM-CoT	aug-train	.25/.29	**.62/.61**	.69/.56	**.72/.67**	.74/.62	.57/.44
SIM-CoT	test	.10/.12	**.44/.38**	.50/.28	**.56/.44**	.54/.32	.32/.19
SIM-CoT-3B	test	.09/.12	.25/.21	**.54/.51**	.55/.31	**.22/.22**	.24/.15
Faithfulness ridge / MLP (alarm confirmed by behaviour)							
GPT-2	aug-train	.00/.00	**.19/.17**	.00/.00	**.40/.36**	.00/.00	.00/.00
GPT-2	test	.00/.00	.09/.08	.00/.00	**.29/.31**	.00/.00	.00/.00
CODI-1B	aug-train	.02/.01	.01/.00	.02/.02	**.36/.36**	.05/.05	.00/.00
CODI-1B	test	.04/.05	.06/.06	.00/.00	**.34/.35**	.02/.02	.00/.00
SIM-CoT	aug-train	.02/.02	**.21/.20**	.02/.01	**.28/.29**	.03/.03	.00/.00
SIM-CoT	test	.03/.03	**.26/.30**	.03/.01	**.26/.28**	.02/.04	.00/.00
SIM-CoT-3B	test	.06/.02	.09/.12	**.37/.36**	.02/.03	**.20/.23**	.00/.00

6 Results: four uses of the map

6.1 A monitor can read the prompt instead of the plan

Table 5 shows that a probe on *any* thought, position 0 and the unused last thought included, decodes whether the problem is multi-step at 0.76–0.95 balanced accuracy and the chain's first operator at 0.50–0.87 against 0.25. The best slot for the operator is never a carrier: it is the question register or a store position. Only the computed value separates the slots. A monitor evaluated on a property the prompt already fixes (a trigger word, a task type, a difficulty) can therefore score well while reading nothing the model computed. For a trigger-in-the-prompt backdoor evaluation this is a confound, and the fix is cheap: run the same probe on a prompt-only control at the same slot.

Table 5: Balanced accuracy of a probe decoding two prompt-fixed properties from each thought (ridge / MLP), standard profile. Chance 0.25 (operator) and 0.50 (long chain).

Model	Property	0	1	2	3	4	5	best / carriers
GPT-2	operator	.63/.76	.55/.66	.63/.74	.60/.67	.62/.76	.60/.68	4 / {1,3}
GPT-2	long chain	.81/.91	.81/.86	.84/.92	.79/.84	.85/.91	.80/.84	2 / {1,3}
CODI-1B	operator	.85/.86	.85/.87	.81/.84	.72/.72	.74/.73	.74/.75	1 / {3}
CODI-1B	long chain	.89/.91	.92/.93	.91/.94	.91/.91	.90/.92	.92/.92	2 / {3}
SIM-CoT	operator	.84/.80	.71/.71	.73/.74	.74/.73	.74/.75	.72/.72	0 / {1,3}
SIM-CoT	long chain	.93/.95	.93/.94	.93/.95	.93/.93	.92/.94	.91/.93	0 / {1,3}

6.2 Thoughts do not catch errors better than confidence

Table 6 is a negative result for the application and a positive one for the map. Nothing read from the thoughts predicts a wrong answer better than the answer's own first-token confidence (0.89–0.92 AUROC), not even 12,288 raw state features, and adding the carrier readout to confidence gains under 0.01. But on the 1B and 3B models the error signal inside the thoughts sits at the carriers the swaps found and nowhere else (carrier 0.82 vs inert 0.60 on CODI-1B; at 3B the two carriers are the best single slots, 0.78 and 0.76, against 0.55–0.66 elsewhere), an independent confirmation of the placement from a task it was never fitted to. On GPT-2 the uncertainty is in every slot.

Table 6: AUROC for "the native answer is wrong" on the full GSM8K test set (1,319 rows). †The 3B card has no inert slot; its weak slots 0 and 1 stand in.

Model	Wrong	Confidence	Carriers	Inert	All slots	Raw states
GPT-2	746	0.889	0.850	0.853	0.851	0.859
CODI-1B	587	0.912	0.821	0.596	0.827	0.835
SIM-CoT	584	0.913	0.754	0.650	0.799	0.836
SIM-CoT-3B	502	0.915	0.793	0.688†	0.807	0.844

6.3 A value-matched write into the carrier

Table 7 shows repair rates. With the needed value from the gold chain, transplanting the nearest same-value natural thought into the carrier turns 14–23% of failures into the gold answer, against 1.7–3.6% for a different-value donor; the intervals do not overlap. At the 1B and 3B models the same transplant harms only 5–9% of correct answers; at GPT-2, 19%. Two preregistered predictions were wrong in useful ways: the centroid of same-value states repairs as well as a retrieved natural one, so retrieval buys nothing over averaging; and adding the store position's K/V *lowers* repair, because the donor's K/V carry the donor's problem as well as the value. With the value from the model's own explicit chain plus a calculator, repair survives at CODI-1B (18.8%, 83% of the gold rate) and SIM-CoT-3B (10.6%, 75%) and weakens at the other two (9.2%, 8.4%). At 3B the second carrier (thought 4) repairs as much as the first (16.3% vs 14.2% with the gold value) and harms twice as many controls (16%), and the absolute writer edges the borrowed vector for the first time (17.2%, overlapping intervals). On every model the explicit chain read directly is right on more of the same failures (18.6–22.7% at 1B and below, 39% at 3B) than injection recovers, so this is a mechanism result: the audited thought is a writable channel, confirmed by two independent sources of the value, not a method that beats reading the chain.

Table 7: Repair rate (%) on wrongly answered $\geq$ 3-step GSM8K test problems whose carrier read the wrong value and that have a donor; harm (%) on correctly answered controls. 95% bootstrap intervals for the main column.

Value from	Model	n	Nearest donor	Random same	Centroid	Different value	Store K/V	State+K/V	Harm
gold chain	GPT-2	357	15.1 [11.5, 19.0]	17.6	20.2	3.6	11.5	12.3	18.7
gold chain	CODI-1B	266	22.6 [17.7, 27.8]	21.8	23.7	2.6	10.2	13.2	7.0
gold chain	SIM-CoT	273	15.4 [11.4, 19.8]	9.9	13.2	2.6	4.0	4.4	5.2
gold chain	SIM-CoT-3B	239	14.2 [10.0, 18.8]	12.1	15.9	1.7	6.3	6.7	8.6
calculator	GPT-2	282	9.2 [6.0, 12.8]	9.2	12.4	3.5	7.1	7.1	16.8
calculator	CODI-1B	208	18.8 [13.9, 24.0]	14.4	16.8	3.8	10.1	10.6	6.0
calculator	SIM-CoT	214	8.4 [5.1, 12.1]	7.0	8.9	3.3	2.3	3.3	10.4
calculator	SIM-CoT-3B	207	10.6 [6.8, 15.0]	9.2	12.6	3.4	4.8	6.3	7.8

6.4 Route on confidence: the cheap gain

The one policy that raised accuracy on every CODI model was switching to the model's own explicit chain, but as run it generates the chain on every problem. Section 6.2 showed that the answer's first-token confidence is the best error detector available, so we used it as a router: the per-problem confidence from the error-detection runs was joined, by question, with the explicit chains from the repair runs on the same 838 long-chain test problems, and a problem is sent to its chain only when its latent confidence falls below a threshold. Sweeping the threshold gives net accuracy against the fraction of problems routed (Table 8).

With a quarter of problems routed the gain is 3.7 points at CODI-1B and 7.5 at SIM-CoT-3B, more than any thought write obtained at any cost (Table 11). With half routed the cascade beats running the chain everywhere (7.6 vs 5.3 at CODI-1B, 11.1 vs 9.7 at 3B): the chain is wrong on some problems the confident latent pass had right, and the gate keeps it away from them. Half of the full-switch gain arrives by 16–23% routed on the 1B and 3B models and by 49% on GPT-2, whose confidence separates its errors less well. The oracle row, routing exactly the wrong problems, shows the headroom (10–17 points); closing it is the error-detection problem, which the thoughts do not solve. The margin gate lands within 1.3 points of the probability gate everywhere, and neither is uniformly better. Costs here are counted in problems routed, not tokens, and the chain is the model's own teacher-mode chain with a calculator correction.

Table 8: Confidence-gated cascade: net accuracy change (points) on the 838 long-chain GSM8K test problems when the least-confident fraction of problems is routed to the model's own explicit chain. 95% row-bootstrap intervals at 25% and 50%. The last row is the best thought-write policy from Table 11.

Routed	GPT-2	CODI-1B	SIM-CoT	SIM-CoT-3B
10%	+0.8	+0.8	+0.5	+2.7
25%	+1.5 [−0.2, +3.2]	+3.7 [+1.9, +5.1]	+1.9 [−0.3, +4.3]	+7.5 [+5.1, +9.9]
50%	+3.1 [+0.1, +5.7]	+7.6 [+4.5, +9.7]	+4.9 [+1.6, +7.8]	+11.1 [+7.5, +14.3]
100% (always explicit)	+5.6	+5.3	+3.2	+9.7
Peak (at % routed)	+7.1 (74%)	+7.6 (50%)	+5.6 (64%)	+11.3 (52%)
Oracle (exactly the wrong ones)	+12.0 (64%)	+11.5 (50%)	+10.3 (50%)	+17.2 (44%)
Best gated thought write	+1.0	+3.4	+1.1	+0.9

7 Results: Coconut is the opposite case

Table 9 runs the same audit on Coconut. The predicted pairing from its curriculum (two thoughts per step) appears exactly: thought 0's state and position 1's K/V transport at the same rate, as do 2 and 3, and 4 and 5. Everything else inverts. Coconut transports on short chains (0.45 at the first thought), so the chain-length gate is a CODI property. Its long-chain transport is 0.52, higher than any CODI model. Its last position is in use: the state is discarded, but its K/V transport 0.15 and hiding them costs 14 points. Every pair is necessary: removing a single thought costs 45–49 points. And it cannot be written to: a same-value donor repairs 5.6% against 3.1% for a different value, while harming 29% of correct answers, because a Coconut thought transplanted from another problem carries that problem. Same six slots, same task, opposite monitoring profile.

Table 9: Coconut (GPT-2 retrain, gate 34.57%). Transport on short (standard, $n = 748$) and long ($n = 1{,}024$) chains; masking cost on long chains; repair on 360 primary targets.

	s_0	s_1	s_2	s_3	s_4	s_5	k_1	k_3	k_5
transport, short	**.45**	.07	.03	.01	.02	.00	**.45**	.03	.02
transport, long	.19	.12	**.52**	.10	.14	.00	.19	**.52**	.15
masking cost, long (pts)	49 (mean)	48 (mask)	45 (mean)	46 (mask)	16 (mean)	14 (mask)			

repair: nearest same-value donor 5.6%, different value 3.1%, centroid 9.4%; harm 29% (donor state), 88% (whole tail)

8 Discussion

What this means for monitoring latent reasoning. For the CODI family a monitor can be placed where it means something, at the carrier, and the audit finds that slot in about an hour. Placing it at the most readable thought is a mistake, since the most readable thoughts are routinely the ones the behaviour never confirms, and any thought will happily report the prompt. For Coconut-style models the picture is reversed: every even thought is both used and needed, and readability tracks use. The practical rule is to run the audit before trusting a probe, and to include a prompt-only control in every monitor evaluation.

Can the repair raise accuracy? A little, and only on one model; we tried three ways to push it further and report all three. We costed policies on the 838 long-chain test problems using the per-target outcomes of the calculator runs (Table 11). Applying the one-slot repair only when the model's explicit chain disagrees with its latent answer (a trigger that needs no answer key) gains 3.4 points on CODI-1B, 1.1 on SIM-CoT, 0.4 on SIM-CoT-3B and nothing on GPT-2. The obvious alternative, switching to the explicit chain's answer on the same disagreement, gains 5.3, 3.2, 9.7 and 5.6 points and needs no transplant.

A learned writer (a ridge or MLP map from a number's token embedding to a carrier vector, fitted on the 2,048 bank thoughts) needs no donor and covers 85–87% of failures instead of 73–80%, with a random-value control at 1–4%; yet its repair rate is level with the borrowed vector (23.2% vs 21.4% on CODI-1B, overlapping intervals) and its gated gain is lower (+2.2 to +2.8). *Writing every thought that holds a wrong step* (located by a slot-to-step map estimated from the bank) is level with or below the one-slot write on all three models and harms more correct answers (14–23% of touched controls). The limit is not the writer.

Where the limit is (Table 10). Classifying each failure from the model's own explicit chain and a calculator shows that 58–68% are plan errors at 1B and below, where no correct value exists to write, and 4–9% are arithmetic done at the answer position, after the thoughts. Only the latent-only failures, 19–23% of the total, are reachable, and the one-slot write already repairs 20–38% of those. So thought correction tops out at three to four points on these models and stays below reading the explicit chain. We then asked whether a stronger model moves its errors into the thoughts, and ran the same kernel on SIM-CoT-3B. It does: the latent-only class nearly doubles, to 39%, and plan errors fall to 48%. But the thoughts do not become more writable: the one-slot write repairs 13% [6.5, 20.8] of the 3B model's latent-only failures, a third of the 1B rate, the ladder's random control is level with its real writes on that class, and the gated policy gains 0.4 points. The value is spread over two carriers, and writing either or both reaches the same few failures. What rises with scale is the worth of the model's own explicit chain, which is right on 39% of its failures and gains 9.7 points when read on disagreement, and Section 6.4 shows that a confidence gate gets more than that gain at half the explicit cost. What the repair shows is a property of the model, a writable channel at a known slot, and what the taxonomy shows is how much of a latent reasoner's error lives in its thoughts at all.

Why the two families differ. CODI distils a teacher into thoughts through a projector and drops the chain's last step during training; what emerges is a narrow channel that holds one intermediate at one slot, with the question kept in position 0's memory and a backup route around the thoughts in the larger model. Coconut replaces steps with thoughts by a curriculum, step by step; what emerges is a thought sequence that is the computation, dense and necessary. Step supervision (SIM-CoT) moves CODI part of the way toward the Coconut profile: more necessary thoughts and a heavier reliance on the question register. Scale moves it further along the same line: at 3B there are two

Table 10: Failure classes among wrong multi-step GSM8K test answers, from the model's own explicit chain plus a calculator, and the one-slot repair rate per class (calculator-supplied value, targets with a mismatched carrier and a donor).

Model	Plan wrong	Latent-only	Upstream arithmetic	Final step
GPT-2 (512 failures)	68% / 2%	19% / 24%	9% / 3%	4% / 5%
CODI-1B (422)	58% / 2%	23% / 38%	9% / 11%	9% / 3%
SIM-CoT (418)	62% / 1%	21% / 20%	11% / 9%	6% / 4%
SIM-CoT-3B (366)	48% / 0%	39% / 13%	8% / 0%	5% / 0%

Table 11: Net accuracy change (points) on the 838 long-chain GSM8K test problems under three policies, scaled from the repair runs' outcomes (calculator-supplied value). Latent accuracy: GPT-2 35.6, CODI-1B 49.6, SIM-CoT 50.1, SIM-CoT-3B 56.3.

Policy	GPT-2	CODI-1B	SIM-CoT	SIM-CoT-3B
One-slot donor, every problem	-1.8	$+2.8$	-2.0	-1.8
One-slot donor, only when the explicit chain disagrees	-0.1	$+3.4$	$+1.1$	$+0.4$
Learned writer (residual), on disagreement	$+1.0$	$+2.2$	$+0.7$	-0.4
Ladder writes to every mismatched thought, on disagreement	$+0.4$	$+1.7$	-0.9	$+0.9$
Switch to the explicit chain's answer on disagreement	$+5.6$	$+5.3$	$+3.2$	$+9.7$

carriers, the gate is gone, the terminal position is in use and the question register matters a third as much, though the backup route around the thoughts remains.

Limitations. All results are on grade-school arithmetic; ASDiv transfers only the short-chain and terminal findings because its multi-step subset is small, and we found no further dataset with equation chains and small numbers. The Coconut checkpoint is a third-party retrain at GPT-2 scale. The repair's needed value came from the gold chain or from the model's own chain; a deployable repairer would need a tool that supplies it, and reading that chain directly is already better. Outcomes are first answer tokens for the CODI family, with native decoding for headline conditions. The cascade's cost is counted in problems routed, not tokens, and GPT-2's batched decoding is not bit-stable, so 39 of its native answers differ between the two runs it joins. Two probe families were tried, not more.

Errata we made along the way. We first gated the CODI-1B reproduction against 51.9%, which is an ablation row in its paper; the main figure is 55.6% and the checkpoint matches it. We first wrote that C4 held on all three training-pool cards; it held on one. We first read GPT-2's cost difference between pools as a regime change; the intervals overlap. All three are recorded in the ledger with the correction.

9 Conclusion

Hidden thoughts in CODI-style models are optional, narrow channels: they carry a value through one or two trained slots, only on hard problems below 3B, they mislead readers, and a value aimed at the carrier takes, though writing them is worth less than reading the model's own explicit chain, and more so as the model grows; routing the least confident quarter of problems to that chain is the one accuracy gain here that is both large and cheap. Coconut-style thoughts are necessary, entangled and uneditable. A cheap, inference-only audit tells you which kind you have, where a monitor would mean something, and what a monitor's accuracy could be measuring instead.

More errata. Three further predictions failed after the first draft: a learned writer did not raise repair above the borrowed vector, writing every mismatched thought did not raise it above one slot, and a stronger model (SIM-CoT-3B) moved its errors into the thoughts without making them more repairable. All three are reported above rather than removed.

Reproducibility statement

The repository `github.com/0x0shephard/latent-reasoning` contains the audit driver, every runner, 80 unit tests for the audit code, the pinned Kaggle notebooks, 15 cards, the per-target outcomes of every repair and error-detection run, and the ledger (`docs/RESEARCH_CONTEXT_LEDGER.md`, sections 108–131) with each preregistration and its scoring. Checkpoints are pinned by revision and SHA-256. Each card records the GPU, library versions and dataset hash. Five reruns, on another account and day, reproduced the twin predictions byte for byte.

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